

Executive Summary

Overview

This report presents the findings of an exploratory data analysis conducted on a warehouse inventory management dataset. The analysis aims to identify inefficiencies, problems and provide actionable insights to improve operations.

Objective

The primary objective is to evaluate inventory accuracy, operational efficiency, and process effectiveness across multiple warehouses.

Methodology

- **Univariate Analysis:** -
Firstly, I performed the univariate analysis on every column of the tables and identify the purpose of particular column.
- **Bivariate Analysis:** -
After finishing my Univariate analysis, I performed a detailed Bivariate analysis. By performing Bivariate analysis, I got to know the relationships between two columns of the table.
- **Multivariate Analysis:** -
After finishing Bivariate analysis on every table, I performed Multivariate analysis across all the tables to finding a deeper relation between more than 2 columns of dataset.

Key Findings

- Some warehouses have more discrepancies
- Incomplete tasks increase errors
- Validation failures lead to mismatches

Business Impact

These problems can cause:

- Inventory loss / mismatch
- Higher costs
- Operational inefficiency
- Slower movement

Recommendations

- Improve validation process
- Optimize task allocation
- Monitor high-risk warehouses time to time